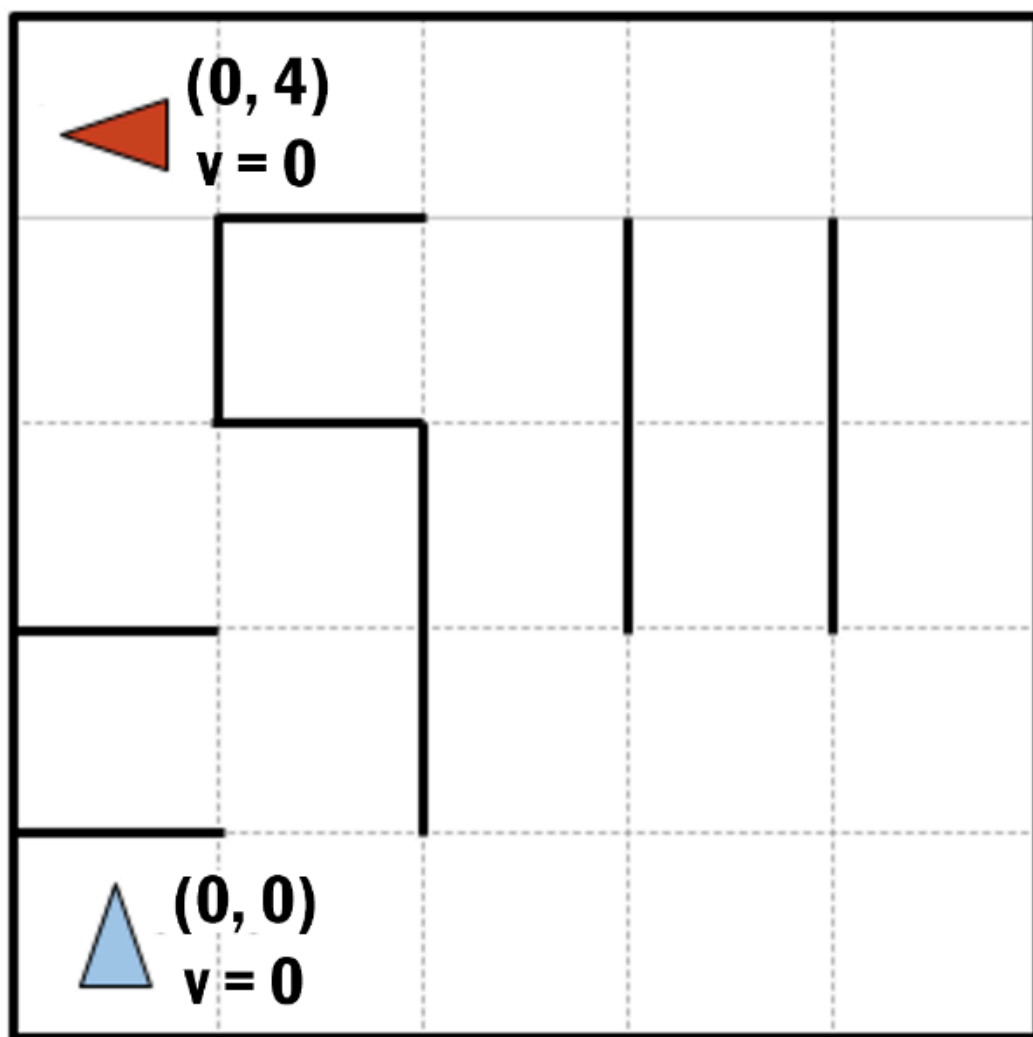


Q1.



Move forward/backward at adjustable velocity v

- ↳ Fast increments v by 1
- ↳ Slow decrements v by 1
- ↳ Maintaining leaves v as is.

Turn Left & Right (move 90°) when $v=0$

Any action that crashes the agent, increasing v above v_{max} or decreasing below v_{min} is NOT allowed

Goal: Find a plan that which parks it (stationary) on exit square using as few actions as possible.

1. $v=1$

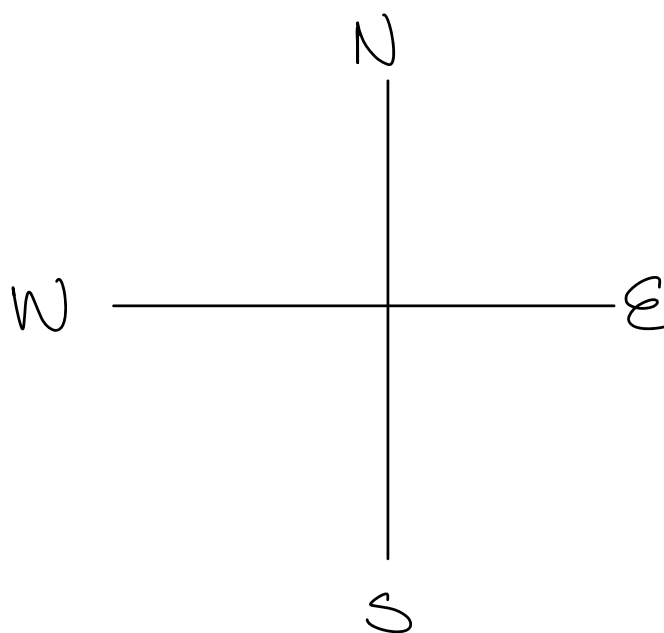
Turn right

Move forward fast

Move forward slow

Turn left

Move forward fast



Move forward maintain
 Move forward slow
 Turn left
 Move forward fast
 Move forward slow
 Turn right
 Move forward fast
 Move forward maintain
 Move forward slow
 Turn left

V2:

Turn right
 Move forward fast
 Move forward maintain
 Move forward slow
 Turn left
 Move forward fast
 Move forward fast
 Move forward slow
 Move forward slow
 Move forward slow
 Turn left
 Move forward fast
 Move forward maintain
 Move forward slow

2.

$$4 \times (M \times N) \times (V_{\max} + 1)$$

3.
 We will have maximum branching if we are at a block from where can go anywhere

Now, the set of moves from here will depend on value of v

If $v = 0$:
 Turn left, Right, Move forward fast

$$b = 3$$

If $v = 1$:
 Move forward fast, Move forward slow, Move forward maintain

$$b = 3$$

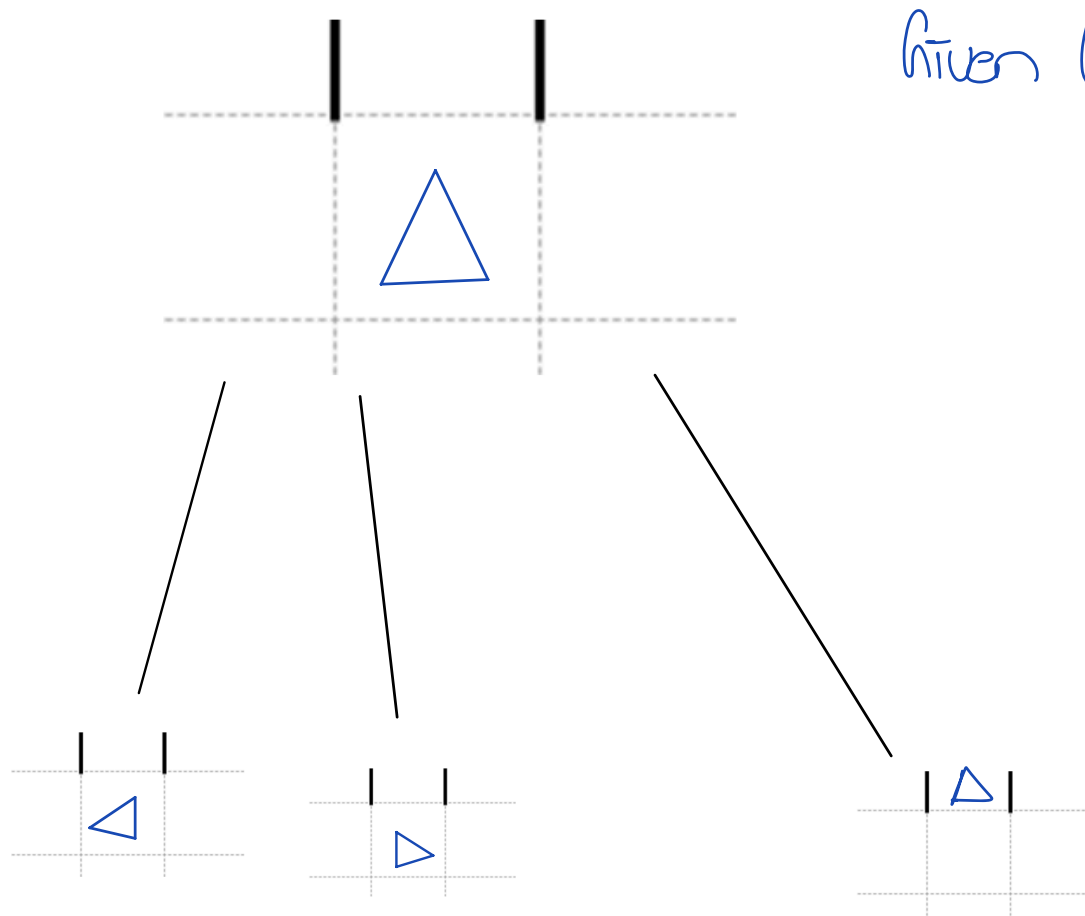
If $1 < v < V_{\max}$:

Move forward fast, Move forward slow, Move forward maintain

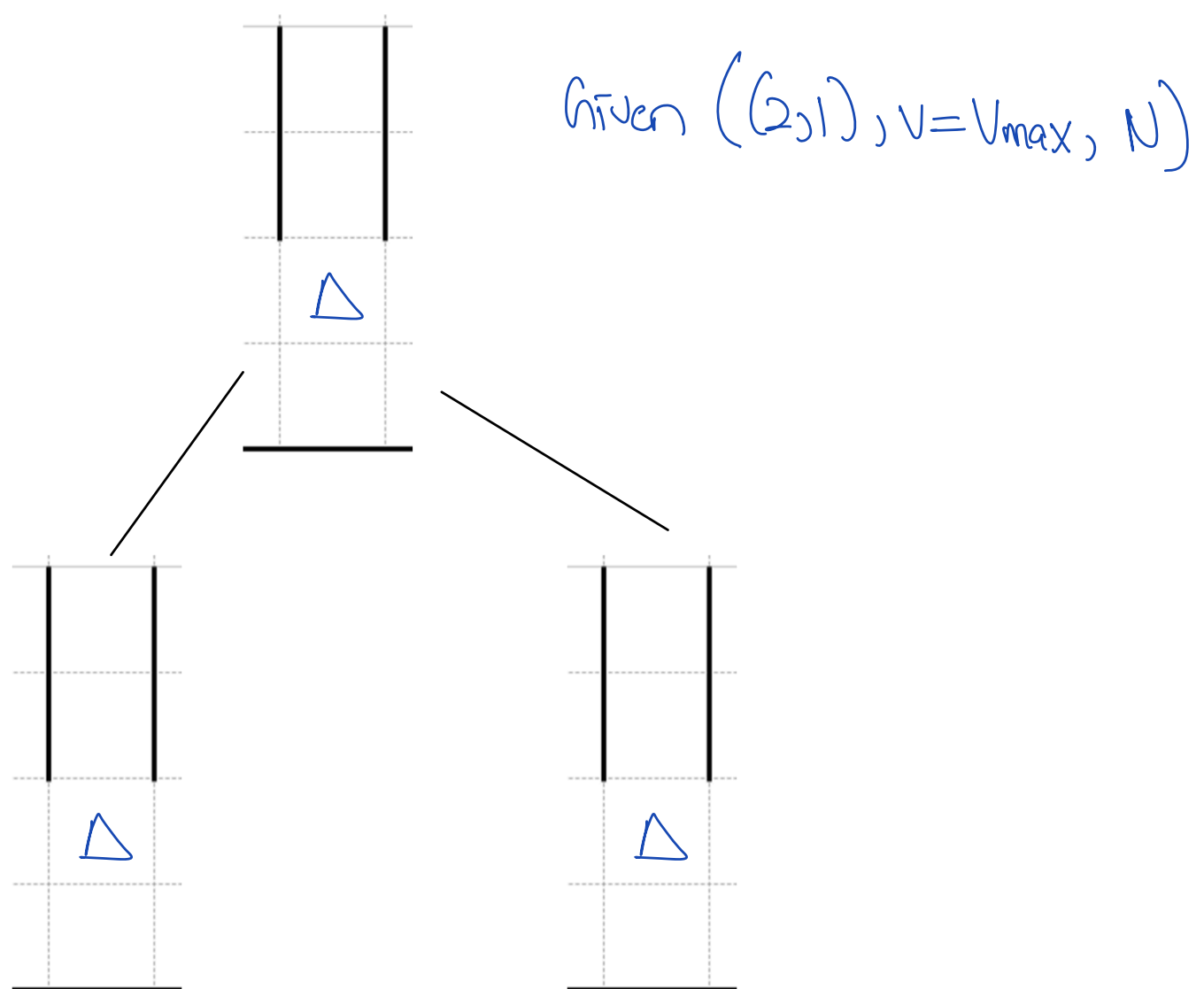
$$b = 3$$

If $V = V_{\max}$
 Move forward slow, Move forward maintain $b=2$

So,
 our maximum branching factor is 3



4.
 following the logic in 3, minimum branching factor is 2



Q2
Constraints:

Move one ring at a time

Never place larger ring on top of a smaller ring

1. I would represent this as a $M \times N$ grid. For simplicity, the agent will in a fixed velocity.

So,

we will have

$(x, y), direction, ring-loaded?, rings-sorted-home)$

↓
current
location

↓
current
direction

↓
boolean
represents
if agent is
carrying a
ring

↓
Rings that are
in right order
at designated
research
station

2.

Given R rings,

$N \times M \times 4 \times 2 \times R+1$

↓
num
locations

↓
direction

↓
carrying
ring

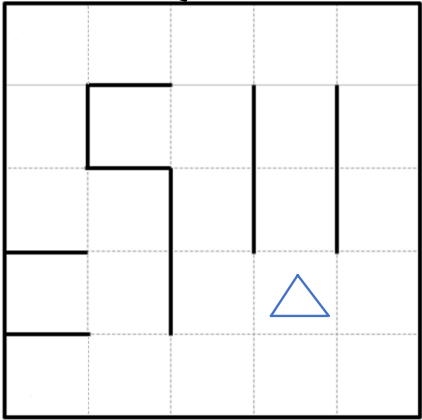
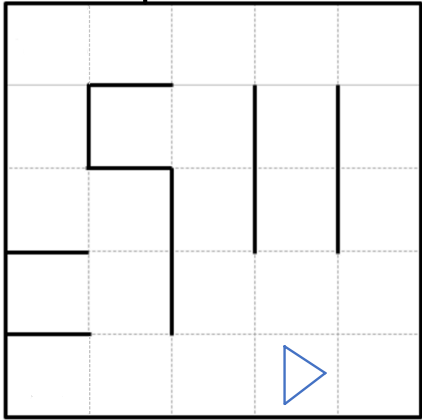
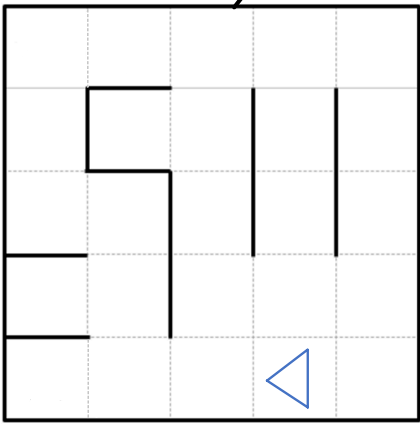
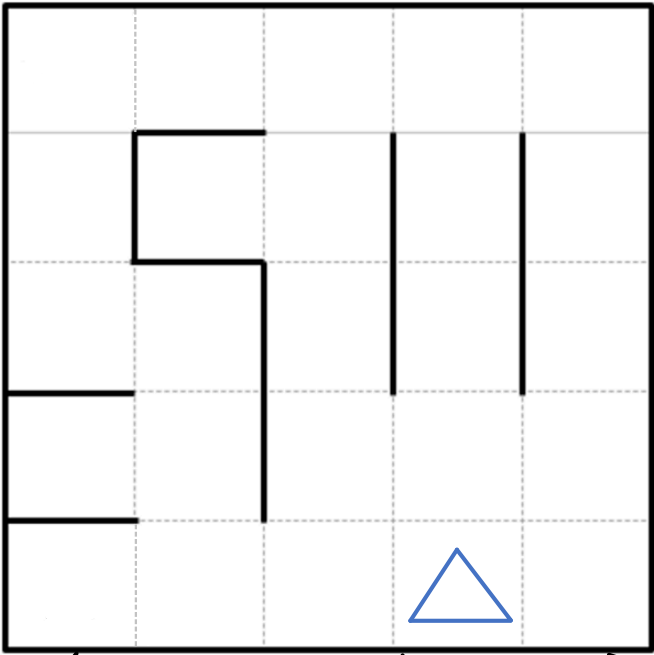
→ No. of rings at home

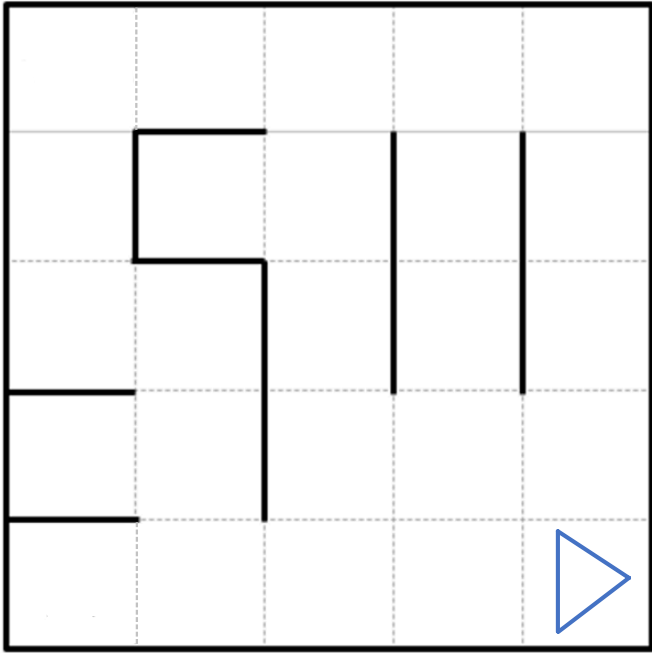
3.

$((0,0), N, 0, 0)$

4. Move forward, Turn left, right, load/unload a ring

5. If rings-sorted-home is equal number of rings then this is our goal state.





velocity > 1